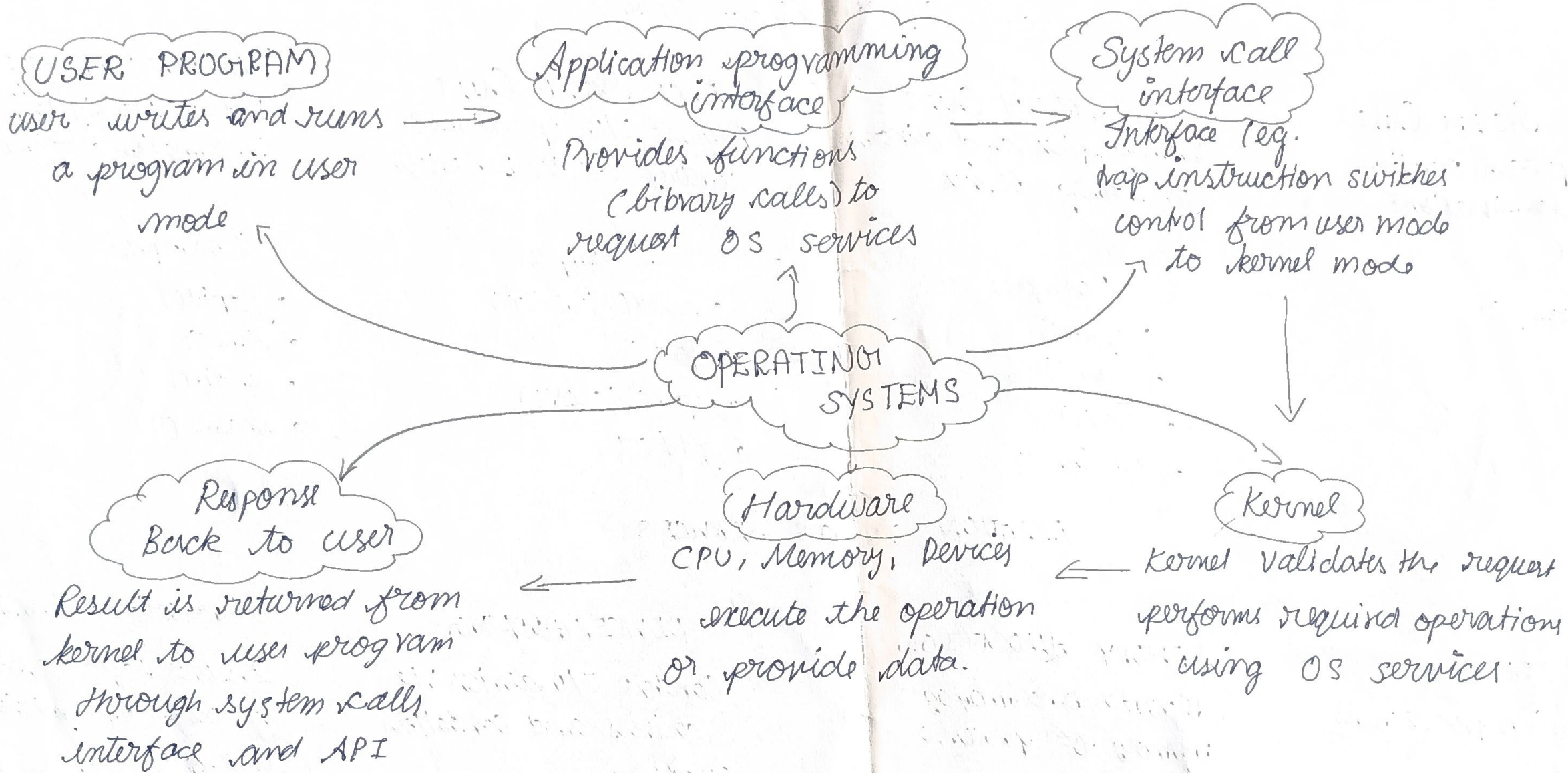


EXECUTION FLOW CONCEPT MAP

Shows how a USER Program request is executed using System Calls from user mode to kernel



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CSA0404 - Operating
Systems

System calls act as a bridge allowing user programs to request OS services

SYSTEM CALLS

Interface between User Programs and Operating System Services

PROCESS SYSTEM CALLS

Manage process creation, execution, termination & control

control ↓

Examples

- fork()
- exec()
- exit()
- wait()

FILE SYSTEM CALLS

Manage files and directories on storage devices

Examples

- open()
- read()
- close()
- unlink()

DEVICE SYSTEM CALLS

Request I/O operations & control device behaviour

Examples

- read()/write()
- ioctl()
- open()/close()
- get/set

INFORMATION SYSTEM CALLS

Get or set system information & attributes

Examples

- getpid()
- time()
- chdir()
- uname()

RELATION TO OS SERVICES

SCHEDULING

Decides which process runs next

Used by:

Process system calls.

MEMORY ALLOCATION

Allocates and manages memory for processes

Used by:

All system call types

DEVICE HANDLING

Controls I/O devices via drivers and controllers

Used by:

Device system calls

Protection & Security

Ensures safe access to resources & permission

Used by: All system call types